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Learning the miRNA→gene regulatory matrix M (design note)

Status: design / parked. Not in any spine. Companion to MODELING_FRAMEWORK.md (the fixed-M pressure model), FORMULAS.md, EDGE_QUESTION_TAXONOMY.md, GENE_QUESTION_TAXONOMY.md, and PRESSURE_FUTURE_OPTIONS.md.

Origin: a mathematician’s framing of the edge as a matrix M mapping a miRNA abundance vector x to a gene-expression vector y . This note records the correct mathematical object, what it would add **over both prior literature and our own fixed-M model**, and exactly where the prior, the confounders, and the many-to-many structure (seed-family overlap, cooperative proximity) enter.

0. The object, stated precisely

$X \in \mathbb{R}^{\{S \times A\}}$	samples $\times$ miRNA-arms	($x[m,s] = \log_2(\text{RPM}+1)$, our spine input)
$Y \in \mathbb{R}^{\{S \times G\}}$	samples $\times$ target genes	(target expression)
$C \in \mathbb{R}^{\{S \times K\}}$	samples $\times$ confounders	(CPE, HRD, target-CN, prolifer, batch)
$M \in \mathbb{R}^{\{G \times A\}}$	genes $\times$ arms	(the regulatory matrix = our edge_w)

$$Y = -X \cdot M^T + C \cdot B + E$$

$M_{\{g,m\}} \geq 0$ is the repressive strength of arm m on gene g (sign carried by the leading $-$). **This is not new to us — it is our model written down.** Our $\text{pressure}(g,s) = \sum_m \text{expr_mult}(m,s) \cdot \text{edge_w}(m,g)$ is row g of $-X \cdot M^T$ evaluated at sample s (MODELING_FRAMEWORK.md §4.1). The only difference the mathematician proposes: **we currently import M from miR-TarBase/TargetScan; he proposes to learn it from the (X, Y) pairs.**

Two corrections to the framing he offered (recorded so they are not re-litigated):

- **Not bilinear in (x, y).** y is the outcome, not an input. The honest name is **multivariate multiple regression**; with a low-rank constraint on M , **reduced-rank regression (RRR)**. Bilinearity re-enters legitimately only in
 - a low-rank factorization $M \approx U \cdot V^T$ (bilinear in latent comodule factors — this is the Theia/PiMiM/NMF family) and (b) a feature model for M ’s *entries* $M_{\{g,m\}} = \varphi(g)^T \theta_{\psi(m)}$ (bilinear in gene-features $\times$ miRNA-features — the factorization-machine reading of TargetScan-style scoring), and (c) **cooperative product terms** (§4), which are genuinely bilinear in x .

- **Not an HMM.** A Markov/HMM needs a state sequence (time or trajectory). TCGA is cross-sectional. The only Markov structure we have is the coarse 3-state GTEX→NAT→tumor chain, and we already model it as *state-differenced* $M \cdot x$ (MODELING_FRAMEWORK.md §8), not as learned transitions. HMM is the wrong tool here unless pseudotime/longitudinal data appears.

Nesting property (the single strongest selling point): freeze every coefficient at its prior and the learned model collapses *exactly* to the current fixed-M pressure. So the learned model is a strict generalization — fixed-M is the infinite-penalty / point-mass-prior limit. Anything it finds is “the prior, plus what the data insists on adding.”

1. Where the prior goes — not just non-zeroneess

This is the central question, and the answer is that a curated edge can inform the estimate in **four** distinct places, only the first of which is “support”:

#	Channel	What enters	Mechanism
1	Support / inclusion	<i>which</i> $M_{\{g,m\}}$ may be nonzero	hard mask to HE edges, or spike-and-slab inclusion prob $\pi_{\{g,m\}} = f(\text{evidence})$
2	Penalty weight (magnitude shrinkage)	seed quality, assay-directness, evidence counts	adaptive/weighted L1: $\lambda_{\{g,m\}} = \lambda / w_{\text{prior}}(g,m)$. Strong edge → small penalty → free to be large; weak edge → heavy shrinkage
3	Prior mean (location)	the curated edge weight itself	Bayesian $M_{\{g,m\}} \sim N(\mu_{\{g,m\}}, \tau^2)$, $\mu_{\{g,m\}} = \text{our edge}_w$. Pulls toward the literature magnitude, not just toward 0
4	Sign / box constraint	repression directionality	$M_{\{g,m\}} \geq 0$ (non-negativity), encoding “miRNAs repress” as a hard constraint

So yes — the prior on seed quality, Functional-MTI evidence counts/quality, reporter/protein vs binding-only directness, and ENCORI CLIP depth all enter as the magnitude prior (channels 2-3), not merely as a 0/1 mask. This is exactly our confidence_logclass numerator and edge_w denominator (MODELING_FRAMEWORK.md §2.3) repurposed: the quantity we currently *use as the fixed weight* becomes the *prior mean and inverse penalty* of the learned weight. TargetScan context++ and miRGeneDB guide/passenger status enter the same way (and the bilinear feature model in §0(b) is the way to let them inform entries we have no curated evidence for — the orphan extension).

This is the first differentiator over prior work. The LASSO miRNA-target papers (Lu 2011) use sequence prediction only for channel 1 (support). GenMiR++ uses sequence + a flat target indicator. None of them push a *quality-weighted, assay-directness-graded* evidence score into channels 2-3. We already have that score, non-circular and breast-audited.

2. Where the confounders go — the matrix view’s cleanest gift

C·B is an additive block in the same regression (MODELING_FRAMEWORK.md §6.2 confounders: CPE, HRD, target-CN, the three proliferation proxies, sequencing batch). **M is only identified after projecting C out of both X and Y** — i.e. M is estimated in the residual space ($I - P_C$). This is precisely our partial-Spearman ladder, done jointly and on all entries at once instead of one edge at a time on ranks.

Restated for the mathematician in his language: a confounder that drives both a miRNA and a gene (proliferation does) creates **omitted-variable bias** in M unless it is in the design. That single sentence is our entire §6.2, and it is the **second differentiator**: essentially none of the prior expression-network methods adjust for tumor purity / proliferation / target-CN / batch. Composition confounding is the #1 failure mode of these networks and we control it by construction.

Two ways to run it, both honest: - **Conditioned (regression)**: put C in as unpenalized covariates. Bias-corrects for *measured* confounders. - **Instrumented (causal)**: use the **miRNA-locus copy number as an instrument** for arm abundance (§5). Corrects for *unmeasured* confounders too. We already compute miRNA-locus CNV (mirna_locus_cnv.py) and have never used it as an instrument — only target-CN as a confounder. This is the one genuinely new estimator the matrix view unlocks on our existing data.

3. Seed-family overlap → collinearity → group structure (identifiability)

Paralogous arms sharing a seed (miR-29a/b/c, miR-200 family, miR-17~92 members) hit the same sites and co-vary — we have measured this (within-family arm $\rho \approx 2\times$ between-family, hubs 20-37% redundant; seed_family_justification, MODELING_FRAMEWORK.md §4.4). In a *joint* regression that documented caveat becomes a hard numerical problem: **near-collinear columns of X**. Plain LASSO will arbitrarily keep one family member and zero the rest, unstably across folds — the non-identifiability re-surfacing as variable-selection instability.

The fix turns the caveat into a **modeling constraint**, which is the **third differentiator**:

- **Group LASSO at the seed-family level** — all arms of a seed family form one group, in-or-out together; magnitude shared/distributed within the group. This literally implements “read share at the seed-family level as the safe unit; the within-family arm is a nomination.”
- **Or collapse the family to one predictor** (mean/max/sum) — our existing family-coupling rung (MODELING_FRAMEWORK.md §6.3, mir301_family_depth).

Either way, the estimand is the **family→gene** weight; the per-arm split is explicitly a within-group attribution, not an identified quantity. Per-edge methods that ignore this report unstable single-arm coefficients as if identified.

4. Co-targeting vs cooperative proximity — two opposite phenomena, don't conflate

The question lumps “same/overlapping seed” with “synergy when in close proximity.” They pull in **opposite directions** and need different terms:

Phenomenon	Site relationship	Effect on repression	Model term	Prior that triggers it
Seed redundancy / competition	same or overlapping site	sub-additive (substitutable; can't both bind)	group structure (§3) + optional per-site saturation	shared seed / overlapping site coords
Cooperativity	<i>distinct</i> sites in close proximity (~8–40 nt)	super-additive / synergistic	product interaction + $M_{\{g, (m, m')\}} \cdot x_m \cdot x_{m'}$	TargetScan site coordinates within a window

Key points:

- A plain linear $\sum_m M_{\{g, m\}} x_m$ is **purely additive** — it cannot represent either synergy or saturation. Cooperativity needs an explicit **bilinear product term in x** (this is the only place a product of two *miRNA abundances* is biologically correct — and it answers the mathematician's “bilinear” instinct with a real mechanism, distinct from the latent-factor bilinearity of §0).
- **The prior decides the interaction support.** We do not fit all $A \cdot (A-1)/2$ products (hopeless, $p \gg n$). We add a product term only for arm pairs with *distinct* predicted sites within the cooperativity window on the same 3'UTR — drawn from TargetScan site coordinates. So the prior is doing double duty: defining the *main-effect* support (§1) **and** the *interaction* support.
- **Global saturation we already model**, separately: the AGO/RISC capacity gate (`ago_gate.py`) is a cohort-level saturating nonlinearity. Local same-site competition (§3 sub-additivity) is a *different*, per-site effect and would need its own grouping/saturation — flag, do not assume the AGO gate covers it.

This pair (redundancy as group structure; cooperativity as primed product terms) is the **fourth differentiator** and is, as far as the literature scan found, not done in the expression-based miRNA-network methods.

5. Scope: gene-focused vs program-wise vs universe-wise

All three are the same M , restricted to different (genes $\times$ arms) blocks. They trade interpretability against statistical strength:

Lane	Block	Structure	Use it for	Cost
Gene-focused	one gene g , its regulators $R(g)$	one penalized regression (one row of M)	deep single-driver story (PTEN ~91 arms, GATA3, ESR1, ZEB1) — the cleanest validation targets	loses cross-gene strength
Program-wise (recommended primary)	one Hallmark’s genes + their regulators	small M block, optional within-program low-rank (comodules)	native unit; interpretable as “regulatory matrix of EMT/P53”; ties to architecture layer (§5)	moderate
Universe-wise	all genes $\times$ all arms	RRR (low-rank) + group LASSO + interaction priors	structural roll-up, NMF comodules, borrows strength	heaviest; identifiability bites hardest; circularity risk highest

Recommendation: **program-wise as the primary lane** (matches the Hallmark unit, tractable $p \gg n$, directly interpretable), **gene-focused** for the driver/orphan deep dives, **universe-wise** only as the low-rank structural overlay (it is essentially our NMF universe layer, MODELING_FRAMEWORK.md §5.6, applied to M rather than to pressure).

6. The causal/perturbation lane (the instrument)

Stage 1: $x_m = \gamma \cdot \text{CN_locus}(m) + C \cdot \delta + u$ (miRNA-locus CN $\rightarrow$ arm abundance)
Stage 2: $y_g = -M_{\{g,m\}} \cdot \hat{x}_m + C \cdot \beta + \varepsilon$ (2SLS / MR estimate of the causal weight)

$\text{CN_locus}(m)$ is a *natural perturbation* of the miRNA dose that is (largely) not caused by the target’s expression — so it breaks the confounding loop that regression alone cannot. Prior art: CNV $\leftrightarrow$ expression causal inference; miRNA Mendelian-randomization. **We have the instrument already built** (`mirna_locus_cnv.py`, `MIRNA_CNV_DOSAGE_REPORT.md`) and have only ever used CN as a *confounder on the target side*. Caveats to honor: weak-instrument (many miRNA loci have little CN variance — report first-stage F), and locus CN can hit neighboring genes (pleiotropy) — restrict to focal segments. This is the strongest answer to “is the weight causal,” and almost no expression-network paper carries an internal instrument.

7. What we have over prior work — summary

Relative to GenMiR++, the LASSO miRNA-target networks (Lu 2011), Theia/PIMiM comodule methods, and the matrix-factorization miRNA-disease line:

1. **Prior at magnitude, not just support** (§1) — quality-weighted, assay-directness-graded, non-circular, breast-audited evidence enters as prior mean + adaptive penalty. The learned model **neests** our fixed- M model.

2. **Confounder-correct by construction** (§2) — purity/proliferation/target-CN/batch in the design; the #1 failure mode of these networks, controlled.
 3. **Seed-family identifiability as a constraint** (§3), not an ignored collinearity.
 4. **Redundancy vs cooperativity modeled with priced interaction terms** (§4).
 5. **A built-in causal instrument** (miRNA-locus CN, §6).
 6. **Program-wise lane on a curated program atlas** (§5) — every M block reads as a named Hallmark’s regulatory matrix.
 7. **Out-of-distribution validation at the protein layer** (CPTAC L1b, MODELING_FRAMEWORK.md §7) and in independent cohorts — most methods stop at mRNA.
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8. The non-negotiable guardrail: circularity

Our central principle (MODELING_FRAMEWORK.md §0.3 #2) is that the prior never touches the expression it is tested against. **If we learn M from the same (X, Y) we later test coupling on, we destroy that.** The resolution is standard:

- **Sequence/curation prior fixes the support** (which entries can be nonzero) and the prior mean; **expression estimates the magnitudes.**
 - **Validate out-of-sample:** patient-level CV folds (we already do 5-fold, held_out_tuning), and on an **orthogonal layer** the fit never saw — CPTAC protein (L1b) and/or the CN-instrument estimate.
 - Keep the **learned M and the curated M side by side** and report where they disagree (the PTEN→miR-19 vs miR-17 kind of disagreement) as a *finding*, never silently let the data overwrite the prior.
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9. Suggested build order

1. **Gene-focused, conditioned, group-LASSO** on a hub panel (PTEN, GATA3, ESR1, ZEB1, IRF1) — adaptive penalty from edge_w, C unpenalized, seed-family groups. Cheapest; directly comparable to the existing per-edge coupling and the P_others experiment (E4/MH-78). Decision gate: does the joint estimate recover the curated drivers and beat raw-abundance coupling out-of-fold?
2. **Add cooperative product terms** (§4) on that panel, support primed by TargetScan site proximity. Gate: do any survive the penalty + improve held-out fit?
3. **Program-wise** with within-program low-rank (comodules) on 3–5 Hallmarks (EMT, P53, G2M, IFN γ , HYPOXIA). Gate: comodules interpretable; couples on held-out.
4. **CN-instrument (2SLS)** on the same hub panel — report first-stage F, compare causal weight sign/magnitude to the conditioned estimate.
5. **Universe-wise RRR** only if 1–4 pay off — as the structural/NMF overlay.

Each gate mirrors PRESSURE_FUTURE_OPTIONS.md discipline: promote a lane only on a pre-named held-out / orthogonal-layer win, never on in-sample fit.

Related

Path	Content
MODELING_FRAMEWORK.md §2–6	the fixed-M pressure + coupling this generalizes a single entry of M = the edge question (coupling-invariance lemma)
EDGE_QUESTION_TAXONOMY.md	

Path	Content
GENE_QUESTION_TAXONOMY.md	a row-sum of M = the gene-aggregate question ("construction is consequential")
PRESSURE_FUTURE_OPTIONS.md	upstream evidence-scoring lanes that feed the prior (\$1)
mirna_locus_cnv.py / MIRNA_CNV_DOSAGE_REPORT.md	the instrument (\$6)